

COUNTING HALLUCINATIONS IN DIFFUSION MODELS

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ABSTRACT

Diffusion probabilistic models (DPMs) have become a general-purpose family of generative models for data modalities ranging from images to videos, yet they can still produce visually plausible samples (*hallucinations*) whose content violates factual regularities, such as an implausible duplicate cup floating beside another cup. Because such errors may arise wherever generated structure is expected to respect countable facts, the absence of practical methods for systematic quantification limits progress on both diagnosing current models and designing future generative systems under factual constraints. We study this broader problem through a concrete and measurable instance, which we term **counting hallucination**: the generation of an incorrect number of instances or structured parts, such as a hand image with six fingers, even when such patterns are absent from the relevant training distribution. To support controlled evaluation across increasing visual complexity, we construct **CountHalluSet**, a dataset suite with explicit counting criteria, comprising ToyShape, SimObject, and RealHand. On this suite, we define a standardized protocol for quantifying counting hallucinations and use it to examine how common DPM sampling conditions, including solver type, ODE solver order, sampling steps, and initial noise, influence count-based factuality. We further analyze how these failure rates relate to common evaluation metrics, such as Fréchet Inception Distance (FID), showing that widely used perceptual metrics do not consistently capture counting hallucinations.

1 INTRODUCTION

Diffusion probabilistic models (DPMs) (Ho et al., 2020; Dhariwal & Nichol, 2021; Rombach et al., 2022), commonly referred to as diffusion models, now serve as a broadly applicable generative framework, producing high-quality samples in settings that include text-to-image generation (Ho & Salimans, 2021; Rombach et al., 2022), audio synthesis (Kong et al., 2021), video generation (Ho et al., 2022), 3D object generation (Luo & Hu, 2021; Vahdat et al., 2022), and protein structure prediction (Jumper et al., 2021; Abramson et al., 2024). Across these uses, however, a recurring challenge is that models may not faithfully preserve factual properties present in the training data, yielding outputs that conflict with real-world regularities. This issue is especially visible when generated objects contain an incorrect number of spatially organized components, such as extra human fingers (Narasimhaswamy et al., 2024) or additional animal limbs (Borji, 2023).

Recent studies have begun to characterize such failures in text-guided diffusion models from several complementary perspectives. For instance, Borji (2023) systematized qualitative shortcomings in text-to-image models, providing a taxonomy of failure patterns. In parallel, Liu et al. (2024) proposed adversarial search for uncovering undesirable behaviors in text-to-image generation, while Wu et al. (2023) used an iterative procedure to discover and mitigate spurious correlations.

More recently, *hallucination* has been recognized as a distinct diffusion-model failure mode (Aithal et al., 2024). Compared with text-prompt failures, which are often assessed through condition-output alignment (e.g., CLIP score (Hessel et al., 2021) and VQA score (Huang et al., 2025; Zarei et al., 2024)), or with overt degradations such as severe distortion, blur, and artifacts, hallucinated samples (*hallucinations*) can look plausible while violating factual consistency with the data distribution. Such violations include incorrect object counts, geometries that appear natural but are physically impossible, and other breaches of physical regularities, and they need not be tied to any particular conditioning signal. These fine-grained distinctions between valid samples and hallucinations make reliable detection difficult for current AI systems. For example, even state-of-the-art large vision-

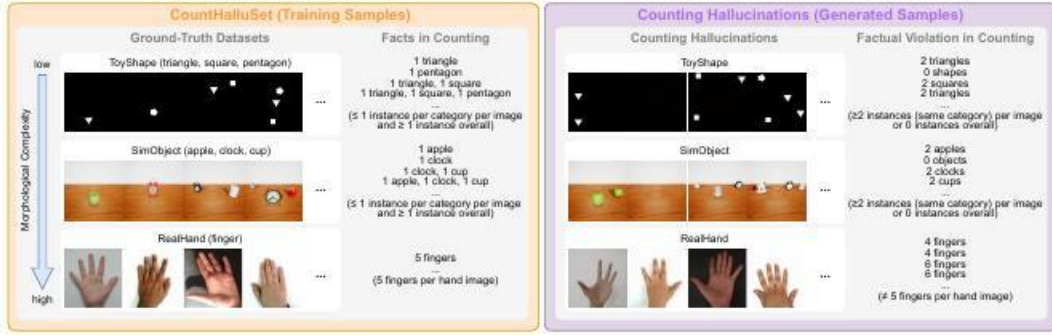


Figure 1: Illustrative comparison between reference samples from the proposed **CountHalluSet** (left) and corresponding model-generated **counting hallucinations** (right). The three datasets instantiate explicit counting criteria over different visual domains: ToyShape and SimObject allow at most one instance per category and require at least one instance overall, whereas RealHand requires exactly five fingers. Counting hallucinations are generated images that violate these dataset-level count constraints, such as two apples in one image or an object-free image with a normal background, even though these patterns never appear in the SimObject dataset.

language models like GPT-4o still struggle with basic tasks such as accurately counting fingers in a clear hand image. A standardized and well-defined protocol for quantifying hallucinated samples is therefore useful beyond any single dataset: it can support principled diagnosis of current models and guide the development of future generative systems with stronger factual consistency, a capability that is central to their use as world models (Ha & Schmidhuber, 2018; Ding et al., 2024; Bruce et al., 2024).

Hallucinations can involve many factual attributes, including counts, colors, spatial relationships, geometry, and physical realism. We focus on counting errors because they provide a tractable entry point into this broader class: count constraints are explicit, visually interpretable, and directly measurable. This motivates the notion of **counting hallucination**. *In this setting, the number of objects in training data is explicitly predetermined. Consequently, a generated image is deemed hallucinated whenever its object counts deviate from these predefined values.* For example, an image depicting a hand with six fingers is classified as a counting hallucination because such a pattern never occurs in the RealHand dataset, as shown in the last row of Fig. 1. To examine the phenomenon across visual regimes rather than a single toy case, we construct CountHalluSet, a dataset suite spanning a spectrum of morphological complexity for countable objects: ToyShape (*triangle, square, pentagon*), SimObject (*mug, apple, clock*), and RealHand (*finger*). Figure 1 illustrates representative reference samples from these datasets together with generated samples that violate their count constraints.

A central requirement for this protocol is reliable identification of the relevant countable objects or parts. Accordingly, we use counting models tailored to the visual structure and label space of each dataset. With these datasets and counters, we evaluate how denoising conditions that are commonly varied in diffusion-model inference—including sampling steps, initial noise, ODE solver order, and solver type—affect counting hallucination rates. We also ask whether standard perceptual metrics track this count-based notion of factuality by correlating counting hallucination rate with Fréchet Inception Distance (FID) Heusel et al. (2017) and MUSIQ Ke et al. (2021).

Our main findings and contributions are:

- Common numerical improvements in diffusion sampling, such as increasing sampling steps from 25 to 100, reduce counting hallucinations in simpler synthetic domains but do not reliably do so for morphologically complex human-hand generation.
- Counting hallucination rate is only weakly and inconsistently correlated with perceptual metrics, whereas non-counting failure rate is strongly correlated with those metrics, indicating that count-based factuality and perceptual quality capture different aspects of generation.
- Guided by the observed role of structural complexity, we propose a training paradigm termed joint-diffusion models, which reduces counting hallucinations and non-counting failures on the RealHand dataset by jointly modeling RGB images and structural masks.

2 RELATED WORK

2.1 HALLUCINATIONS IN DIFFUSION MODELS

Motivated by the broader literature on hallucination in large language models (LLMs) (Kadavath et al., 2022; Ji et al., 2023; Huang et al., 2023), researchers have begun to investigate analogous failures in diffusion-generated samples, including in unconditional settings. For example, Aithal et al. (2024) proposed the qualitative hypothesis that hallucinated samples may emerge from interpolation between distinct data modes. Although such work offers insight into possible mechanisms, diffusion-model hallucinations still lack systematic quantitative tools that apply across controlled generative settings.

This gap motivates our effort to make hallucinations in diffusion-generated outputs more identifiable and measurable. A quantitative lens can clarify how undesirable behavior changes under common modeling and sampling choices, including variations in sampling steps, solver order, and other denoising conditions that affect accumulated generation error.

2.2 QUANTITATIVE METRICS IN IMAGE GENERATION

Visual Question Answering (VQA) scores (Huang et al., 2025; Zarei et al., 2024) and CLIP scores (Hessel et al., 2021) are widely used to evaluate text-guided image generation. However, because these metrics rely on language-based models, their suitability for detecting hallucinated visual content is limited. One reason is that pretrained language models have a statistical propensity to hallucinate certain fact types, which bounds the reliability of detection systems based only on such models (Kalai & Vempala, 2024). Another is the persistent gap between human annotators and LLMs when classifying hallucinated textual content (Lin et al., 2022; Li et al., 2023).

Beyond language-mediated scores, image generation is commonly assessed with quantitative metrics such as FID (Heusel et al., 2017), IS (Salimans et al., 2016), Precision & Recall (Sajjadi et al., 2018), and MUSIQ (Ke et al., 2021). These metrics summarize perceptual or distributional properties through feature embeddings, including those extracted from a pretrained Inception-v3 model (Szegedy et al., 2016). Whether such metrics also reflect the severity of factual hallucinations in generated datasets remains unclear. We address this question for the count-based case, where factual violations can be explicitly defined and measured.

3 PRELIMINARY: DIFFUSION PROBABILISTIC MODELS

3.1 FORWARD (DIFFUSION) PROCESS

Forward Process. Given observed data $\mathbf{x}_0 \sim q_0(\mathbf{x}_0)$, Diffusion Probabilistic Models (DPMs) (Sohl-Dickstein et al., 2015; Ho et al., 2020) define a *forward process* (a.k.a. *diffusion process*) that progressively maps structured data toward noise by adding Gaussian perturbations over T discrete timesteps. At each timestep t , the marginal distribution of the noisy sample $\mathbf{x}_t$ conditioned on the original data $\mathbf{x}_0$ follows:

$$q_{0:t}(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t \mid \sqrt{\bar{\alpha}_t}\mathbf{x}_0, (1 - \bar{\alpha}_t)\mathbf{I}), \quad \alpha_t := 1 - \beta_t, \quad \bar{\alpha}_t := \prod_{j=1}^t \alpha_j, \quad (1)$$

where $t \in \{0, 1, \dots, T\}$ and $\{\beta_i\}_{i=1}^T$ is a pre-defined time-dependent noise schedule, commonly chosen as a linear (Ho et al., 2020) or cosine (Nichol & Dhariwal, 2021) schedule. The factor $\bar{\alpha}_t$ controls the signal-to-noise ratio at step t , with $\bar{\alpha}_0 = 1$ corresponding to uncorrupted data and $\bar{\alpha}_T \approx 0$ corresponding to nearly pure noise. Equivalently, Eq. (1) can be written as:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t}\mathbf{x}_0 + (1 - \bar{\alpha}_t)\boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (2)$$

For sufficiently large T , the terminal marginal $q_{0:T}(\mathbf{x}_T|\mathbf{x}_0)$ approaches the standard Gaussian distribution $\mathcal{N}(\mathbf{0}, \mathbf{I})$. In widely used DPM implementations (Ho et al., 2020; Nichol & Dhariwal, 2021), T is typically set to 1000.

3.2 REVERSE (DENOISING) PROCESS.

Training Objective. The *reverse process* (a.k.a., *denoising process*) learns to transform the diffused variable $\mathbf{x}_T$ back toward samples from the data distribution. In practice, DPMs train a neural network $\epsilon_\theta(\mathbf{x}_t, t)$, known as the *noise prediction model*, to estimate the injected noise ϵ from $\mathbf{x}_t$ and timestep t . The parameter θ is optimized with the following objective (Ho et al., 2020; Song et al., 2022):

$$\min_{\theta} \mathbb{E}_{t, \mathbf{x}_0, \epsilon} [\pi(t) \|\epsilon_\theta(\mathbf{x}_t, t) - \epsilon\|_2^2], \quad (3)$$

where $\pi(t) > 0$ is a weighting function over timesteps.

Ancestral Sampling. Given noisy data $\mathbf{x}_T$, samples from the original distribution $q_0(\mathbf{x}_0)$ can be generated by reversing the diffusion process with the same number of steps, a procedure commonly referred to as *ancestral sampling*:

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = \mathcal{N}(\mathbf{x}_{t-1} | \frac{1}{\sqrt{1-\beta_t}}(\mathbf{x}_t + \beta_t \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)), \beta_t \mathbf{I}), \quad (4)$$

where $\nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ is the unknown term, referred to as the *score function* (Song et al., 2022).

In practice, DPMs approximate the scaled score function $-\sqrt{(1-\bar{\alpha}_t)} \nabla_{\mathbf{x}_t} \log q_t(\mathbf{x}_t)$ using the learned noise prediction model. This yields the following equivalent sampling update associated with Eq. (3):

$$\mathbf{x}_{t-1} = \mu_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \epsilon, \quad \mu_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{1-\beta_t}} \mathbf{x}_t - \frac{\beta_t}{\sqrt{1-\beta_t}} \frac{1}{\sqrt{1-\bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t). \quad (5)$$

ODE-Based Accelerated Sampling. The ancestral sampler in Eq. (4) includes a stochastic term (Kloeden et al., 1992), but the same diffusion family also admits an associated deterministic process formulated as an ordinary differential equation (ODE), known as the *probability flow ODE* (Song et al., 2022):

$$\frac{d\mathbf{x}(\tau)}{d\tau} = f(\tau)\mathbf{x}(\tau) + h(\mathbf{x}(\tau), \tau), \quad f(\tau) = -\frac{1}{2}\beta(\tau), \quad h(\mathbf{x}(\tau), \tau) = -\frac{1}{2}\beta(\tau) \nabla_{\mathbf{x}(\tau)} \log q_\tau(\mathbf{x}(\tau)), \quad (6)$$

where $\tau \in [T, 0]$. Let $\mathbf{x}(\tau)$ denote the ideal trajectory that satisfies this ODE with the true score function $h(\mathbf{x}(\tau), \tau)$ and starts from ideal diffused data $\mathbf{x}(T)$ drawn from the true diffused distribution $q_T(\mathbf{x}(T))$. Set the discrete time steps as $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$, where $\tau_N = 0$. The variation-of-constants formula gives the exact one-step evolution:

$$\mathbf{x}(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h(\mathbf{x}(u), u) du \quad (7)$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state transition operator for the linear component $\frac{d\mathbf{z}}{d\tau} = f(\tau)\mathbf{z}$. The step size $\Delta t > 0$ determines the number of ODE solver steps, $T/\Delta t$, and larger steps enable accelerated sampling relative to ancestral sampling. Because the linear part $\mathcal{G}(t_2, t_1)$ is available in closed form, most accelerated ODE-based samplers (Song et al., 2021; Liu et al., 2022; Zhang & Chen, 2023; Lu et al., 2022; 2023) concentrate their numerical design choices, such as solver order and number of sampling steps, on approximating the nonlinear integral involving h . Appendix A.1 to A.3 provides additional theoretical details used to contextualize these sampling conditions.

4 FORMAL DEFINITION, DATASETS, AND EVALUATION PROTOCOL FOR COUNTING HALLUCINATIONS

This section formalizes counting hallucinations in a way that can be instantiated whenever a dataset imposes explicit count constraints. We first define counting criteria and the corresponding deviations, then introduce the CountHalluSet datasets as concrete testbeds spanning simple shapes, rendered objects, and real hand images. For each dataset, we state the applicable counting criteria, describe the associated hallucination cases, and specify the standardized quantification protocol.

4.1 COUNTING HALLUCINATIONS: DEVIATIONS FROM THE COUNTING CRITERIA

We define counting criteria at the dataset level and treat counting hallucinations as generated samples that deviate from those predefined constraints while remaining visually suitable for counting.

Counting criteria in a training dataset. Let $\mathcal{D}_{\text{ref}}$ be a training dataset. For any reference sample $\mathbf{x} \in \mathcal{D}_{\text{ref}}$, the number of objects from category c , denoted $N_c(\mathbf{x})$, is constrained to lie in a predefined set $\mathcal{S}_c \subseteq \mathbb{Z}_{\geq 0}$ (e.g., $\{0, 1\}$ or $\{1, 2, 3\}$), so that $N_c(\mathbf{x}) \in \mathcal{S}_c$. We also require that a valid reference sample contain at least one object, i.e. $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$, where C is the set of considered categories.

Counting hallucinations as deviations from the counting criteria. Given a generated sample $\hat{\mathbf{x}}$, we say $\hat{\mathbf{x}}$ exhibits a *counting hallucination* with respect to $\mathcal{D}_{\text{ref}}$ when both of the following conditions hold:

1. **Sufficient visual quality for counting.** The image $\hat{\mathbf{x}}$ must have sufficient visual quality for the background and any present objects to be clearly discernible. This condition separates count-specific factual errors from broader generation failures, such as severe degradations. We define a binary indicator $\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \in \{0, 1\}$, referred to as the *counting-ready indicator*, where 1 denotes that the image is suitable for counting and 0 denotes that it is not. Depending on the domain, this indicator can be obtained from human annotation, from a classifier calibrated against human judgments, or from other automated criteria.
2. **Violation of counting facts.** A violation of counting facts occurs when there exists a category c such that $N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$, or when the image contains no objects (i.e., $\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$) despite the dataset requiring at least one object per sample. In our experiments, a counting model predicts object counts in generated images and provides evidence for these violations.

A counting hallucination (CH) is therefore identified only when the sample is counting-ready ($\mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) = 1$) and its predicted counts violate the dataset criteria ($\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c$ or $\sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0$). Formally,

$$\mathbb{I}_{\text{CH}}(\hat{\mathbf{x}}) = \mathbb{I}_{\text{CRI}}(\hat{\mathbf{x}}) \wedge \left(\exists c : N_c(\hat{\mathbf{x}}) \notin \mathcal{S}_c \vee \sum_{c \in C} N_c(\hat{\mathbf{x}}) = 0 \right).$$

4.2 DATASETS, COUNTING HALLUCINATIONS AND EVALUATION PROTOCOL

The CountHalluSet suite contains three datasets for studying counting hallucinations under progressively richer visual structure: ToyShape, SimObject, and RealHand. Examples from each dataset and their corresponding counting hallucinations are shown in Fig. A1 and A2. Appendix A.4 provides additional statistical summaries.

4.2.1 TOYSHAPE

Dataset description. The ToyShape dataset comprises 30,000 images containing three geometric shape categories: *triangle*, *square*, and *pentagon*. All shapes are white, have an equal area of 120 pixels, and do not overlap within an image. This dataset provides a low-complexity setting in which the counting criteria require each image to contain at most one instance of each shape category and at least one shape in total, as shown in the top-left of Fig. 1. Formally, we have $C = \{\text{triangle}, \text{square}, \text{pentagon}\}$. For each class $c \in C$, we set $N_c(\mathbf{x}) \in \mathcal{S}_c = \{0, 1\}$, subject to the constraint $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$.

Counting hallucinations. Under these criteria, any image generated by a model trained on ToyShape that contains two or more shapes from the same category is counted as a hallucination, such as the image with two pentagons in the top-right region of Fig. 1. Empty images, containing only the black background and no shapes, are also classified as counting hallucinations because they violate the at-least-one-object requirement. Representative examples appear in Appendix A.5.

Counting model used in ToyShape. We use a counting model to identify count errors directly and do not apply a counting-ready indicator, since severe visual degradations are rare in this simple domain. Specifically, we fine-tune a ResNet-50 (He et al., 2016) on more than 400k ToyShape images in which each category appears 0–3 times per sample (i.e., $\mathcal{S}_c = \{0, 1, 2, 3\}$), covering

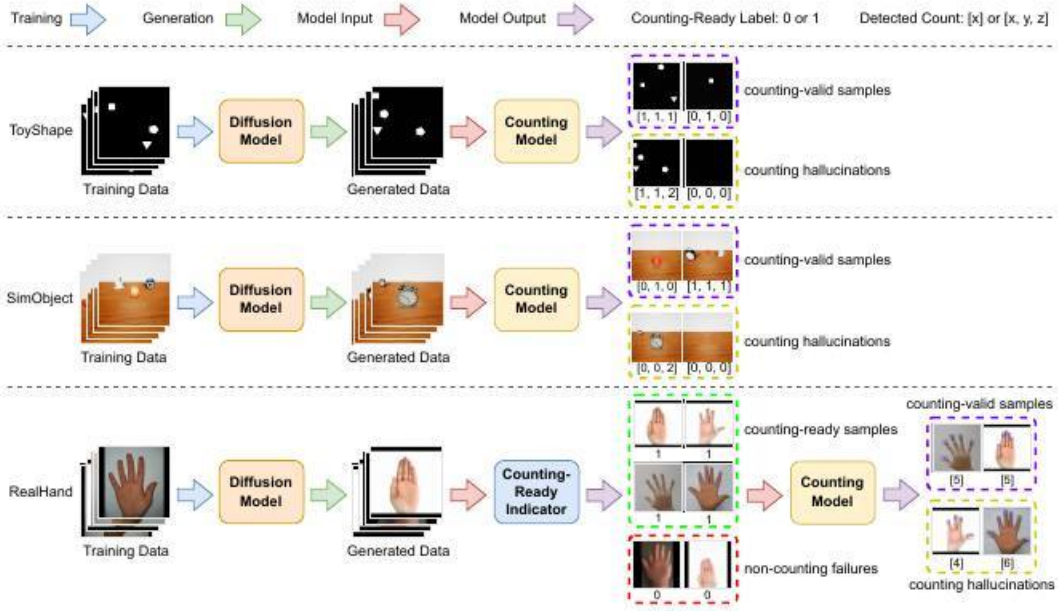


Figure 2: Evaluation procedure for applying the CountHalluSet protocol across datasets. ToyShape and SimObject generated images are assessed directly because severe visual failures are rare in these controlled domains. RealHand adds a counting-ready indicator, implemented as a binary classifier aligned with human judgments, to distinguish counting hallucinations from non-counting failures such as severely deformed fingers. Count labels correspond to ToyShape: *triangle*, *square*, *pentagon*; SimObject: *mug*, *apple*, *clock*; RealHand: *finger*.

both counting-valid samples and plausible counting hallucinations for accurate quantification. The resulting model achieves over 99.9% counting accuracy on generated data.

Quantification of counting hallucinations. For a diffusion model trained on ToyShape, we generate 30,000 samples, matching the training-set size. The counting model then predicts the number of objects in each generated image. Images whose predicted counts violate the ToyShape criteria are identified as counting hallucinations, as illustrated in Fig. 2.

4.2.2 SIMOBJECT

Dataset description. The SimObject dataset contains 30,000 rendered images of everyday objects and introduces more realistic appearance while retaining explicit count constraints. It includes three object categories, *mug*, *apple*, and *clock*, each with 10 distinct intra-class variations. Objects are randomly placed on a wooden tabletop under fixed lighting conditions, with minimal mutual occlusion when any occurs, as shown in the middle-left of Fig. 1. The dataset is synthetically generated in Unreal Engine 5 (Epic Games, 2024), using photorealistic rendering that simulates object geometry, material attributes, lighting, and reflections. The counting criteria are analogous to those of ToyShape (see Sec. 4.2.1). Formally, we have $C = \{mug, apple, clock\}$. For each class $c \in C$, we set $N_c(\mathbf{x}) \in S_c = \{0, 1\}$, subject to the constraint $\sum_{c \in C} N_c(\mathbf{x}) \geq 1$.

Quantification of counting hallucinations. Because SimObject shares the same count-constraint structure as ToyShape, its definition of counting hallucinations, counting-model construction, and quantification protocol follow the same procedure, as summarized in Fig. 2.

4.2.3 REALHAND

Dataset description. To evaluate counting hallucinations in a real-image domain with higher structural complexity, we construct RealHand from 5,050 human hand images sourced from the 11k Hands dataset (Afifi, 2019), Kaggle¹, and Roboflow². We aggregate raw images from these

¹<https://www.kaggle.com/>

²<https://universe.roboflow.com/>

sources and remove images with blur or artifacts. Each retained image depicts a human hand with five fingers, including both dorsal (back) and palmar (front) views under varying lighting conditions and backgrounds (e.g., white surfaces, blackboards, and walls), as shown in the lower-left of Fig. 1. The counting criterion requires exactly five fingers per image, i.e., $C = \{finger\}$ with $N_c(x) \in S_c = 5$ for each $c \in C$, and $\sum_{c \in C} N_c(x) = 5$.

Counting hallucinations and non-counting failures. Realistic hand generation can fail in ways that are not merely count violations, including blurriness, severely deformed fingers, and other visual artifacts in addition to extra or missing fingers. We therefore use a counting-ready indicator, implemented as a binary classifier, to separate images suitable for count evaluation from these broader failures. Appendix A.7 provides examples of both counting hallucinations and non-counting failures.

Counting-ready indicator and counting model used in RealHand. To operationalize this separation, we finetune a MaxViT (Tu et al., 2022) model on 2.5k annotated generated images as a counting-ready indicator aligned with human judgments. For fingertip counting, we finetune a YOLO-12 (Tian et al., 2025) model on 2k generated images with fingertip annotations. In a human evaluation of 500 randomly sampled images by three annotators, the indicator and counting model achieve over 96% and 99% accuracy, respectively. Appendix A.6 shows representative YOLO detection results.

Quantification of counting hallucinations. Unlike ToyShape and SimObject, where all generated images can be directly evaluated because severe degradations are rare, RealHand requires an additional filtering step. We first apply the counting-ready indicator to select hand images with sufficient visual quality, and then use the counting model to predict the number of fingers, as shown in Figure 2.

5 EXPERIMENTS

5.1 IMPLEMENTATION DETAILS

Training of Diffusion Models. We use diffusion backbones matched to the scale and visual complexity of each dataset: a DDPM (Ho et al., 2020) trained from scratch for ToyShape, and latent diffusion models (LDMs) (Rombach et al., 2022), pretrained on CelebA-HQ (Karras et al., 2018), fine-tuned for SimObject and RealHand. The timestep horizon T is set to 1,000 with a standard linear noise schedule. Images are resized to 128×128 for DDPMs and 256×256 for LDMs. All models are trained with the Adam optimizer (Kingma, 2015) at a learning rate of 0.0001 for 150k, 300k, and 80k steps on ToyShape, SimObject, and RealHand, respectively, using an effective batch size of 256.

Inference of Diffusion Models. For ancestral sampling (i.e., DDPM), we use 1,000 sampling steps. For accelerated ODE-based sampling, we compare first- and second-order methods: DPM-Solver-1, equivalent to DDIM (Song et al., 2021), as the first-order solver, and DPM-Solver-2 (Lu et al., 2022) as the second-order solver. Each ODE solver is evaluated under three commonly used denoising-step configurations: 25, 50, and 100 steps. For RealHand, the counting model uses a YOLO detector with a confidence threshold of 0.3 and an IoU threshold of 0.1. Unless otherwise specified, each evaluation generates the same number of samples as the corresponding training set, and reported results are averaged over three random seeds.

5.2 QUANTIFYING COUNTING HALLUCINATIONS UNDER DIFFERENT DENOISING CONDITIONS

Table 1 reports counting hallucination rate (CHR), non-counting failure rate (NCFR), and total failure rate (TFR) under varied denoising conditions, including solver type, ODE solver order, sampling steps, and initial noise, across ToyShape, SimObject, and RealHand. These controlled comparisons reveal how conclusions about sampling quality depend on the complexity of the count-constrained domain. Several key observations follow.

(1) **Increasing sampling steps reduces counting hallucinations in synthetic datasets but can exacerbate them in RealHand.** As shown in Table 1, under the ODE solver settings of 25, 50, and 100 steps, increasing the number of sampling steps generally lowers CHR in ToyShape and SimObject, but tends to raise CHR in RealHand, except at 100 steps with DPM-Solver-2. This contrast indicates that finer solver granularity can help when the data distribution is simple and highly

Table 1: Counting hallucination rates (CHR; the number of counting hallucinations divided by total generated samples) under different solver configurations and initial-noise settings across ToyShape, SimObject, and RealHand. For RealHand, non-counting failure rate (NCFR) captures samples rejected by the counting-ready indicator. Total failure rate (TFR) is the sum of CHR and NCFR. *Diffused* and *Normal* denote ground-truth initial noise and standard Gaussian noise, respectively.

Solver Name	Sampling Steps	CHR (ToyShape)		CHR (SimObject)		CHR (RealHand)		NCFR (RealHand)		TFR (RealHand)	
		<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>	<i>Diffused</i>	<i>Normal</i>
DPM-Solver-1 (Song et al., 2021)	25	2.43	3.47	9.27	9.95	12.71	12.95	16.18	18.06	28.90	31.01
	50	1.83	2.79	8.53	9.27	13.82	13.85	11.32	12.51	25.14	26.36
	100	1.56	2.32	8.04	8.90	14.29	14.55	9.54	10.63	23.83	25.19
DPM-Solver-2 (Lu et al., 2022)	25	2.45	3.81	8.16	8.19	14.23	14.48	8.37	9.33	22.60	23.82
	50	1.76	2.86	7.90	7.95	16.15	15.99	6.38	7.22	22.53	23.21
	100	1.41	2.24	7.89	7.89	15.06	15.43	7.82	8.94	22.88	24.38
DDPM (Ho et al., 2020)	1000	0.63	0.64	4.98	4.99	10.82	10.75	2.07	2.39	12.88	13.14

structured, while morphologically complex real-image domains such as RealHand may expose local inconsistencies that additional denoising steps do not resolve and may even amplify.

(2) **Higher-order ODE solvers reduce non-counting failures but can increase counting hallucinations in RealHand under standard inference settings.** As shown in Table 1, comparing DPM-Solver-1 and DPM-Solver-2 on RealHand at standard sampling budgets such as 25 and 50 steps reveals a consistent trade-off: the higher-order solver substantially reduces non-counting failures while producing a higher CHR. A related pattern appears when the number of sampling steps increases for both DPM-Solver-1 and DPM-Solver-2. Thus, computationally stronger sampling conditions can improve broad visual quality, reflected by lower NCFR, without enforcing factual correctness of finger counts.

(3) **DDPM substantially outperforms the tested ODE solvers in mitigating both counting hallucinations and non-counting failures.** Across these comparisons, ancestral sampling (i.e., DDPM) achieves the lowest CHR, NCFR, and TFR. Within the evaluated settings, DDPM therefore provides a strong reference point for estimating lower failure rates and suggests that stochastic ancestral sampling can be more favorable for reducing count-based and non-counting failures than the tested accelerated ODE solvers.

(4) **A better matched initial noise configuration can reduce both counting hallucinations and non-counting failures.** As shown in Table 1, using “diffused” noise (i.e., ground-truth initial noise) tends to produce lower CHR, NCFR, and TFR than “normal” noise (i.e., standard Gaussian noise). This effect likely arises because diffused noise is better aligned with the initial noise distribution encountered during training, suggesting that informed initialization can mitigate both count-specific factual errors and broader visual failures.

(5) **Greater morphological complexity in countable objects increases the likelihood of counting hallucinations.** Comparing CHR across ToyShape, SimObject, and RealHand shows that diffusion models tend to exhibit higher CHR as the target objects become more morphologically complex. This pattern suggests that fine-grained details and complex spatial relationships make it harder for diffusion models to preserve accurate counts, leading to more frequent additions or omissions in structured objects.

5.3 CORRELATION ANALYSIS BETWEEN COUNTING HALLUCINATION RATE, NON-COUNTING FAILURE RATE, AND PERCEPTUAL METRICS

Because counting hallucination is a count-based form of factual hallucination, a natural question is whether perceptual improvement implies factual improvement: *Does a better perceptual metric, such as a lower FID, necessarily imply fewer counting hallucinations by reflecting a smaller distribution gap between generated and training data?* Our results indicate that the answer is **not necessarily**.

To test this relationship, we analyze correlations between CHR and perceptual metrics, and between NCFR and the same metrics. We consider two widely used measures: FID (Heusel et al., 2017), a distribution-based metric, and MUSIQ (Ke et al., 2021), an instance-based image-quality metric. For correlation estimation, we use three complementary statistics: Pearson correlation (Lee Rodgers & Nicewander, 1988) for linear dependence, Spearman rank correlation (Spearman, 1904) for monotonic

Table 2: Correlations between counting hallucination rate (CHR), non-counting failure rate (NCFR), total failure rate (TFR), and perceptual metrics including FID and MUSIQ. PTI-Hand results are obtained by prompting a pretrained Stable-Diffusion-3.5-Medium model with ten hand-generation prompts to produce 5050 samples, matching the RealHand dataset size. FID for PTI-Hand is computed between RealHand images and the model-generated samples. Here, p denotes the p-value assessing the statistical significance of correlation coefficient r or ρ . $\dagger$ denotes settings that combine DDPM results with results from multiple ODE solvers across sampling conditions.

Metric	CHR vs. FID						NCFR vs. FID						CHR vs. MUSIQ						NCFR vs. MUSIQ					
	SimObject	RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	RealHand	PTI-Hand		RealHand	PTI-Hand	
Pearson (r)	0.87	0.01	-0.97	<0.01	0.20	0.69	0.99	<0.01	0.88	<0.01	0.93	<0.01	-0.02	0.95	-0.97	<0.01	-0.99	<0.01						
Pearson † (r)	0.49	0.14	-0.53	0.13	—	—	0.96	<0.01	—	—	0.46	0.20	—	—	-0.98	<0.01	—	—						
Spearman (ρ)	0.84	<0.01	-0.97	<0.01	-0.14	0.78	0.99	<0.01	0.92	<0.01	0.80	0.01	0.42	0.33	-0.88	<0.01	-0.46	0.29						
Spearman † (ρ)	0.75	0.01	-0.51	0.15	—	—	0.95	<0.01	—	—	0.40	0.28	—	—	-0.91	<0.01	—	—						
Distance Correlation (r_d)	0.89	—	0.97	—	0.47	—	0.99	—	0.91	—	0.93	—	0.42	—	0.97	—	0.99	—						
Distance Correlation † (r_d)	0.82	—	0.74	—	—	—	0.95	—	—	—	0.71	—	—	—	0.97	—	—	—						

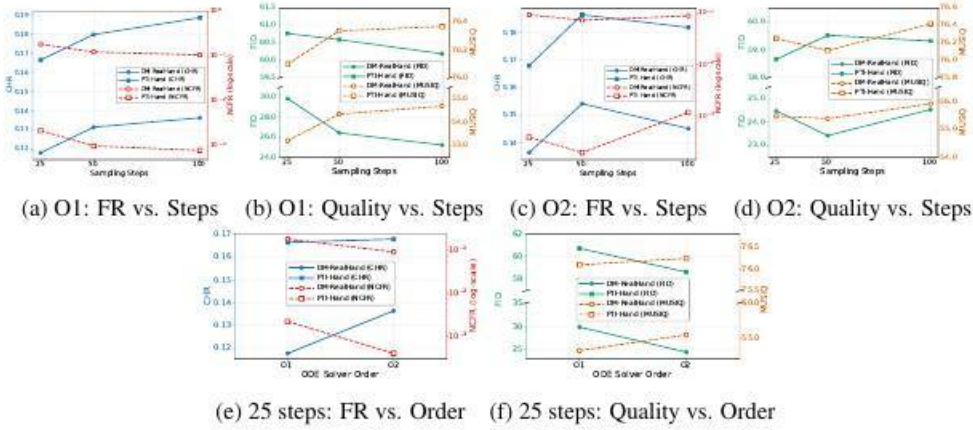


Figure 3: Trend consistency between DM-RealHand (ours) and PTI-Hand (pretrained Stable-Diffusion-3.5-Medium) under comparable sampling changes. O1 and O2 denote first- and second-order ODE solvers (DPM-solver-1 / DPM-solver-2 for DM-RealHand; Euler / Heun for PTI-Hand). FR metrics include CHR and NCFR; quality metrics include FID and MUSIQ. Across variations in sampling steps and solver order, both model settings exhibit highly consistent trends.

dependence, and Distance Correlation (Székely et al., 2007) for general nonlinear dependence. The evidence below shows that metric behavior differs substantially between count-specific factual errors and broader visual failures.

✚ **Correlations between CHR and perceptual metrics are dataset- and solver-dependent rather than intrinsic.** As shown in the first and third rows of Table 2, Pearson correlations between CHR and FID vary substantially across datasets, being positive for SimObject, negative for RealHand, and positive again for PTI-Hand, while the corresponding Spearman correlations shift from positive to negative to negative. Distance correlations also decrease from 0.82 (SimObject) to 0.74 (RealHand) to 0.47 (PTI-Hand), suggesting that more complex generative content weakens the association between CHR and FID. Combining DDPM results with ODE-solver results further attenuates these correlations.

✚ **Correlations between NCFR and perceptual metrics are consistently strong and positive.** In contrast to the weak and unstable correlations between CHR and perceptual metrics, the correlations between NCFR and perceptual metrics remain strong across the evaluated settings. For example, all coefficients for NCFR vs. FID exceed 0.88 and have very small p-values, indicating high statistical significance.

5.4 GENERALIZATION OF REALHAND FINDINGS TO PRETRAINED TEXT-TO-IMAGE MODELS

This section tests whether the RealHand observations extend beyond models trained or fine-tuned in our controlled setup to a large pretrained text-to-image model. Specifically, we evaluate Stable-

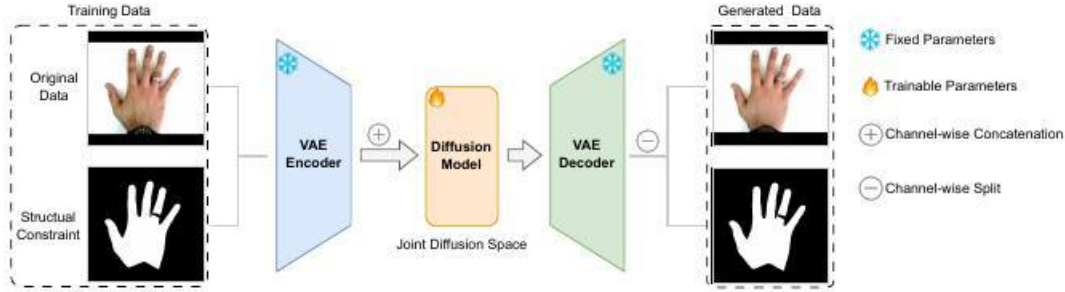


Figure 4: Diagram of the proposed joint-diffusion model (JDM). The model channel-wise concatenates latent representations of hand RGB images and corresponding segmentation masks. This joint learning paradigm supplies explicit structural information during denoising, encouraging anatomically plausible hand poses and correct finger topology, which reduces CHR and NCFR in the evaluated RealHand setting.

Diffusion-3.5-Medium (Esser et al., 2024) with the counting-ready indicator and counting model introduced in Sec. 4.2.3. Appendix A.8 provides additional details. As shown in Figure 3, the effects of solver type and sampling steps on counting hallucinations in this pretrained model align closely with the trends from the controlled RealHand experiments. This agreement supports the validity of the findings in Sec. 5.2 and indicates that the evaluation protocol is not limited to the particular RealHand-trained model. The correlation results in Table 2 provide further support: weak correlations between CHR and perceptual metrics and strong correlations between NCFR and perceptual metrics also appear in PTI-Hand. Together, these results show that the main observations transfer to a pretrained text-to-image setting while remaining grounded in the same hand-generation evaluation criteria.

5.5 HOW TO REDUCE COUNTING HALLUCINATIONS?

The lower counting hallucination rates observed for simpler countable objects (see Observation 5 in Sec. 5.2) suggest that diffusion models preserve count-based factuality more reliably when the relevant structure has low morphological complexity, as in simple object masks. This observation motivates a broader mitigation question: can a diffusion model reduce counting hallucinations by denoising general visual representations together with explicit structural constraints, such as masks that encode coarse topology and counts, within a shared latent space? We refer to models that jointly learn visual representations and structural constraints during denoising as **joint-diffusion models** (JDM).

In our RealHand instantiation, we concatenate the latent representation of each hand mask channel-wise with that of its corresponding RGB image, as shown in Fig. 4. We then finetune the UNet pretrained on CelebA-HQ (Karras et al., 2018), following the RealHand configuration in Sec. 5.1, using RealHand images and their associated masks extracted with SAM-2 (Ravi et al., 2024). To model both modalities jointly, the denoising UNet input and output channels are expanded to match the combined latent dimensionality of the image and mask latents (i.e., $3+3=6$ channels).

As shown in Table 3, JDM achieves lower CHR and NCFR than the vanilla LDM across the evaluated sampling conditions, indicating that explicit structural supervision can help generate count-sensitive, morphologically intricate objects. At the same time, JDM yields higher FID. This degradation likely reflects the added denoising burden of jointly modeling two modalities in a small-scale training regime, creating a trade-off between perceptual fidelity and structural factuality. We also observe promising reductions in counting hallucinations when applying JDM to large-scale pretrained diffusion models under carefully controlled finetuning settings. Appendix A.9 provides further details, results, and discussion.

Table 3: Failure-rate and image-quality comparison between LDM and the proposed JDM on RealHand across ODE solvers and sampling steps. Lower values indicate better performance. Although JDM jointly generates hand RGB images and segmentation masks, FID is computed only on RGB outputs to maintain comparability with the LDM baseline.

Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fréchet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
Sampling Steps	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
LDM	12.95	13.85	14.55	14.48	15.99	15.43	18.06	12.51	10.63	9.33	7.22	8.94	29.86	26.43	25.21	24.48	23.42	24.54
JDM (ours)	11.41	10.67	10.51	10.12	9.66	10.07	2.94	2.42	2.29	3.51	4.63	2.26	37.69	35.51	34.68	34.63	32.07	33.89

6 CONCLUSION

This work develops a systematic framework for quantifying hallucinations in diffusion models through the concrete case of count-based factuality. We introduce **CountHalluSet** and a practical, reproducible protocol for measuring counting hallucinations across simple shapes, rendered objects, and real hand images. Controlled experiments show that numerical sampling strategies that improve or stabilize generation in simpler domains do not reliably reduce counting hallucinations for morphologically complex human hands. Our correlation analyses further show that widely used perceptual metrics, including FID and MUSIQ, do not consistently reflect the severity of counting hallucinations, even though they track non-counting failures more strongly. Experiments with a large pretrained text-to-image model support the broader applicability of the protocol and the main findings within the evaluated hand-generation setting.

REFERENCES

- Josh Abramson, Jonas Adler, Jack Dunger, Richard Evans, Tim Green, Alexander Pritzel, Olaf Ronneberger, Lindsay Willmore, Andrew J Ballard, Joshua Bambrick, et al. Accurate structure prediction of biomolecular interactions with alphafold 3. *Nature*, pp. 493—500, 2024.
- Mahmoud Afifi. 11k hands: Gender recognition and biometric identification using a large dataset of hand images. *Multimedia Tools and Applications*, 78:20835–20854, 2019.
- Sumukh K Aithal, Pratyush Maini, Zachary C Lipton, and J Zico Kolter. Understanding hallucinations in diffusion models through mode interpolation. *arXiv preprint arXiv:2406.09358*, 2024.
- Ali Borji. Qualitative failures of image generation models and their application in detecting deepfakes. *Image and Vision Computing*, 137:104771, 2023.
- Jake Bruce, Michael D Dennis, Ashley Edwards, Jack Parker-Holder, Yuge Shi, Edward Hughes, Matthew Lai, Aditi Mavalankar, Richie Steigerwald, Chris Apps, et al. Genie: Generative interactive environments. In *ICML*, 2024.
- Thomas M Cover and Joy A Thomas. *Elements of Information Theory*. John Wiley & Sons, 2012.
- Prafulla Dhariwal and Alexander Nichol. Diffusion models beat gans on image synthesis. *NeurIPS*, 34:8780–8794, 2021.
- Jingtao Ding, Yunke Zhang, Yu Shang, Yuheng Zhang, Zefang Zong, Jie Feng, Yuan Yuan, Hongyuan Su, Nian Li, Nicholas Sukiennik, et al. Understanding world or predicting future? a comprehensive survey of world models. *ACM Computing Surveys*, 2024.
- Epic Games. Unreal engine, 2024. URL <https://www.unrealengine.com>.
- Patrick Esser, Sumith Kulal, Andreas Blattmann, Rahim Entezari, Jonas Müller, Harry Saini, Yam Levi, Dominik Lorenz, Axel Sauer, Frederic Boesel, et al. Scaling rectified flow transformers for high-resolution image synthesis. In *ICML*, pp. 12606–12633, 2024.
- David Ha and Jürgen Schmidhuber. World models. *arXiv preprint arXiv:1803.10122*, 2(3), 2018.
- Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *CVPR*, pp. 770–778, 2016.

- Jack Hessel, Ari Holtzman, Maxwell Forbes, Ronan Le Bras, and Yejin Choi. Clipscore: A reference-free evaluation metric for image captioning. In *EMNLP*, pp. 7514–7528, 2021.
- Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. Gans trained by a two time-scale update rule converge to a local nash equilibrium. *NeurIPS*, 30, 2017.
- Jonathan Ho and Tim Salimans. Classifier-free diffusion guidance. In *NeurIPS 2021 Workshop on Deep Generative Models and Downstream Applications*, pp. 1–14, 2021.
- Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *NeurIPS*, 33: 6840–6851, 2020.
- Jonathan Ho, Tim Salimans, Alexey Gritsenko, William Chan, Mohammad Norouzi, and David J Fleet. Video diffusion models. *NeurIPS*, 35:8633–8646, 2022.
- Kurt Hornik, Maxwell Stinchcombe, and Halbert White. Multilayer feedforward networks are universal approximators. *Neural networks*, 2(5):359–366, 1989.
- Kaiyi Huang, Chengqi Duan, Kaiyue Sun, Enze Xie, Zhenguo Li, and Xihui Liu. T2i-compbench++: An enhanced and comprehensive benchmark for compositional text-to-image generation. *IEEE TPAMI*, 2025.
- Lei Huang, Weijiang Yu, Weitao Ma, Weihong Zhong, Zhangyin Feng, Haotian Wang, Qianglong Chen, Weihua Peng, Xiaocheng Feng, Bing Qin, et al. A survey on hallucination in large language models: Principles, taxonomy, challenges, and open questions. *arXiv preprint arXiv:2311.05232*, 2023.
- Ziwei Ji, Nayeon Lee, Rita Frieske, Tiezheng Yu, Dan Su, Yan Xu, Etsuko Ishii, Ye Jin Bang, Andrea Madotto, and Pascale Fung. Survey of hallucination in natural language generation. *ACM computing surveys*, 55(12):1–38, 2023.
- John Jumper, Richard Evans, Alexander Pritzel, Tim Green, Michael Figurnov, Olaf Ronneberger, Kathryn Tunyasuvunakool, Russ Bates, Augustin Žídek, Anna Potapenko, et al. Highly accurate protein structure prediction with alphafold. *Nature*, 596(7873):583–589, 2021.
- Saurav Kadavath, Tom Conerly, Amanda Askell, Tom Henighan, Dawn Drain, Ethan Perez, Nicholas Schiefer, Zac Hatfield-Dodds, Nova DasSarma, Eli Tran-Johnson, et al. Language models (mostly) know what they know. *arXiv preprint arXiv:2207.05221*, 2022.
- Adam Tauman Kalai and Santosh S Vempala. Calibrated language models must hallucinate. In *Proceedings of the 56th Annual ACM Symposium on Theory of Computing*, pp. 160–171, 2024.
- Tero Karras, Timo Aila, Samuli Laine, and Jaakko Lehtinen. Progressive growing of gans for improved quality, stability, and variation. In *ICLR*, 2018.
- Junjie Ke, Qifei Wang, Yilin Wang, Peyman Milanfar, and Feng Yang. Musiq: Multi-scale image quality transformer. In *ICCV*, pp. 5148–5157, 2021.
- Diederik P Kingma. Adam: A method for stochastic optimization. In *ICLR*, pp. 1–15, 2015.
- Peter E Kloeden, Eckhard Platen, Peter E Kloeden, and Eckhard Platen. *Stochastic differential equations*. Springer, 1992.
- Zhifeng Kong, Wei Ping, Jiaji Huang, Kexin Zhao, and Bryan Catanzaro. Diffwave: A versatile diffusion model for audio synthesis. In *ICLR*, pp. 1–17, 2021.
- Joseph Lee Rodgers and W Alan Nicewander. Thirteen ways to look at the correlation coefficient. *The American Statistician*, 42(1):59–66, 1988.
- Junyi Li, Xiaoxue Cheng, Wayne Xin Zhao, Jian-Yun Nie, and Ji-Rong Wen. Halueval: A large-scale hallucination evaluation benchmark for large language models. In *EMNLP*, pp. 6449–6464, 2023.
- Stephanie Lin, Jacob Hilton, and Owain Evans. Truthfulqa: Measuring how models mimic human falsehoods. In *ACL*, pp. 3214–3252, 2022.

- Luping Liu, Yi Ren, Zhijie Lin, and Zhou Zhao. Pseudo numerical methods for diffusion models on manifolds. In *ICLR*, pp. 1–24, 2022.
- Qihao Liu, Adam Kortylewski, Yutong Bai, Song Bai, and Alan Yuille. Discovering failure modes of text-guided diffusion models via adversarial search. In *ICLR*, pp. 1–25, 2024.
- Cheng Lu, Yuhao Zhou, Fan Bao, Jianfei Chen, Chongxuan Li, and Jun Zhu. Dpm-solver: A fast ode solver for diffusion probabilistic model sampling in around 10 steps. *NeurIPS*, 35:5775–5787, 2022.
- Cheng Lu, Yuhao Zhou, Fan Bao, Jianfei Chen, Chongxuan Li, and Jun Zhu. Dpm-solver++: Fast solver for guided sampling of diffusion probabilistic models. In *ICLR*, pp. 1–19, 2023.
- Shitong Luo and Wei Hu. Diffusion probabilistic models for 3d point cloud generation. In *CVPR*, pp. 2837–2845, 2021.
- Supreeth Narasimhaswamy, Uttaran Bhattacharya, Xiang Chen, Ishita Dasgupta, Saayan Mitra, and Minh Hoai. Handdiffuser: Text-to-image generation with realistic hand appearances. In *CVPR*, pp. 2468–2479, 2024.
- Alexander Quinn Nichol and Prafulla Dhariwal. Improved denoising diffusion probabilistic models. In *ICML*, pp. 8162–8171, 2021.
- Michael A Nielsen. *Neural networks and deep learning*, volume 25. Determination press San Francisco, CA, USA, 2015.
- Nikhila Ravi, Valentin Gabeur, Yuan-Ting Hu, Ronghang Hu, Chaitanya Ryali, Tengyu Ma, Haitham Khedr, Roman Rädle, Chloe Rolland, Laura Gustafson, et al. Sam 2: Segment anything in images and videos. *arXiv preprint arXiv:2408.00714*, 2024.
- Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *CVPR*, pp. 10684–10695, 2022.
- Mehdi SM Sajjadi, Olivier Bachem, Mario Lucic, Olivier Bousquet, and Sylvain Gelly. Assessing generative models via precision and recall. *NeurIPS*, 31, 2018.
- Tim Salimans, Ian Goodfellow, Wojciech Zaremba, Vicki Cheung, Alec Radford, and Xi Chen. Improved techniques for training gans. *NeurIPS*, 29, 2016.
- Jascha Sohl-Dickstein, Eric Weiss, Niru Maheswaranathan, and Surya Ganguli. Deep unsupervised learning using nonequilibrium thermodynamics. In *ICML*, pp. 2256–2265, 2015.
- Jiaming Song, Chenlin Meng, and Stefano Ermon. Denoising diffusion implicit models. In *ICLR*, pp. 1–22, 2021.
- Yang Song, Jascha Sohl-Dickstein, Diederik P Kingma, Abhishek Kumar, Stefano Ermon, and Ben Poole. Score-based generative modeling through stochastic differential equations. In *ICLR*, pp. 1–36, 2022.
- C Spearman. The proof and measurement of association between two things. *The American Journal of Psychology*, 15(1):72–101, 1904.
- Christian Szegedy, Vincent Vanhoucke, Sergey Ioffe, Jon Shlens, and Zbigniew Wojna. Rethinking the inception architecture for computer vision. In *CVPR*, pp. 2818–2826, 2016.
- Gábor J Székely, Maria L Rizzo, and Nail K Bakirov. Measuring and testing dependence by correlation of distances. *The Annals of Statistics*, pp. 2769–2794, 2007.
- Yunjie Tian, Qixiang Ye, and David Doermann. Yolov12: Attention-centric real-time object detectors. *arXiv preprint arXiv:2502.12524*, 2025.
- Zhengzhong Tu, Hossein Talebi, Han Zhang, Feng Yang, Peyman Milanfar, Alan Bovik, and Yinxiao Li. Maxvit: Multi-axis vision transformer. In *ECCV*, pp. 459–479, 2022.

-
- Arash Vahdat, Francis Williams, Zan Gojcic, Or Litany, Sanja Fidler, Karsten Kreis, et al. Lion: Latent point diffusion models for 3d shape generation. *NeurIPS*, 35:10021–10039, 2022.
- Shirley Wu, Mert Yuksekgonul, Linjun Zhang, and James Zou. Discover and cure: Concept-aware mitigation of spurious correlation. In *ICML*, pp. 37765–37786, 2023.
- Dmitry Yarotsky. Universal approximations of invariant maps by neural networks. *Constructive Approximation*, 55(1):407–474, 2022.
- Arman Zarei, Keivan Rezaei, Samyadeep Basu, Mehrdad Saberi, Mazda Moayeri, Priyatham Kattakinda, and Soheil Feizi. Understanding and mitigating compositional issues in text-to-image generative models. *arXiv preprint arXiv:2406.07844*, 2024.
- Qinsheng Zhang and Yongxin Chen. Fast sampling of diffusion models with exponential integrator. In *ICLR*, pp. 1–33, 2023.

A APPENDIX

A.1 PROOF OF THE BOUND OF TOTAL ACCUMULATED DISTRIBUTIONAL ERROR IN ANCESTRAL SAMPLING

For the joint distributions of the ideal reverse process $q(\mathbf{x}_{T:0})$ and the model reverse process $p_\theta(\mathbf{x}_{T:0})$, the KL divergence is defined as:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0})||p_\theta(\mathbf{x}_{T:0})) = \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_{T:0})}{p_\theta(\mathbf{x}_{T:0})} \right]. \quad (\text{A.1})$$

Using the Markov property, both reverse processes $q(\mathbf{x}_{T:0})$ and $p_\theta(\mathbf{x}_{T:0})$ decompose as follows:

$$q(\mathbf{x}_{T:0}) = q(\mathbf{x}_T) \prod_{t=1}^T q(\mathbf{x}_{t-1} | \mathbf{x}_t), \quad (\text{A.2})$$

$$p_\theta(\mathbf{x}_{T:0}) = p_\theta(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t). \quad (\text{A.3})$$

Applying the chain rule to this decomposition yields:

$$\begin{aligned} \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0})||p_\theta(\mathbf{x}_{T:0})) &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T) \prod_{s=1}^T q(\mathbf{x}_{s-1} | \mathbf{x}_s)}{p_\theta(\mathbf{x}_T) \prod_{s=1}^T p_\theta(\mathbf{x}_{s-1} | \mathbf{x}_s)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_{T:0})} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} + \sum_{t=1}^T \log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \\ &= \mathbb{E}_{q(\mathbf{x}_T)} \left[\log \frac{q(\mathbf{x}_T)}{p_\theta(\mathbf{x}_T)} \right] + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} \left[\mathbb{E}_{q(\mathbf{x}_{t-1} | \mathbf{x}_t)} \left[\log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \middle| \mathbf{x}_t \right] \right] \\ &= \mathcal{D}_{\text{KL}}(q(\mathbf{x}_T)||p_\theta(\mathbf{x}_T)) + \sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t)||p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]. \quad (\text{A.4}) \end{aligned}$$

According to the Data Processing Inequality (Cover & Thomas, 2012), marginalizing over, or equivalently discarding, random variables cannot increase the KL divergence:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0)||p_\theta(\mathbf{x}_0)) \leq \mathcal{D}_{\text{KL}}(q(\mathbf{x}_{T:0})||p_\theta(\mathbf{x}_{T:0})). \quad (\text{A.5})$$

Combining Eq. (A.4) and Eq. (A.5) gives the bound of total accumulated distributional error:

$$\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0)||p_\theta(\mathbf{x}_0)) \leq \underbrace{\mathcal{D}_{\text{KL}}(q(\mathbf{x}_T)||p_\theta(\mathbf{x}_T))}_{\text{diffused-prior gap}} + \underbrace{\sum_{t=1}^T \mathbb{E}_{q(\mathbf{x}_t)} [\mathcal{D}_{\text{KL}}(q(\mathbf{x}_{t-1} | \mathbf{x}_t)||p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t))]}_{\text{accumulated transition errors caused by the model prediction error}}, \quad (\text{A.6})$$

Equality would hold only in the ideal case where $q(\mathbf{x}_T) = p_\theta(\mathbf{x}_T)$ and $p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) = q(\mathbf{x}_{t-1} | \mathbf{x}_t)$ for all t and $\mathbf{x}_t$, meaning that the diffusion model exactly recovers the underlying data distribution $q(\mathbf{x}_0)$. This upper bound on the overall KL divergence ($\mathcal{D}_{\text{KL}}(q(\mathbf{x}_0)||p_\theta(\mathbf{x}_0))$) between true data and model-generated data matches the DDPM variational bound (Ho et al., 2020) when the data entropy $H(\mathbf{x}_0)$ is excluded, suggesting that optimizing the DDPM variational loss tightens an explicit upper bound on the discrepancy between true and generated data.

Diffused-prior gap. The first error source is the mismatch between the distribution of ground-truth diffused data $q_T(\mathbf{x}_T) = q_{0:T}(\mathbf{x}_T|\mathbf{x}_0)$ and the prior distribution used for sampling $p_\theta(\mathbf{x}_T)$. This mismatch exists because the number of diffusion steps T is finite in practice. The discrepancy can be quantified with a distributional metric such as:

$$\mathcal{D}_{\text{KL}}(q_T(\mathbf{x}_T)||p_\theta(\mathbf{x}_T)), \quad p_\theta(\mathbf{x}_T) = \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (\text{A.7})$$

where $\mathcal{D}_{\text{KL}}$ denotes the Kullback–Leibler (KL) divergence. Other appropriate metrics, such as FID or Wasserstein distance, can also be used to evaluate this distributional discrepancy. Because the

diffused-prior gap is present at the initial sampling state, its effect can propagate through the denoising process and contribute to persistent output error.

Noise model prediction error. The second error source comes from the noise prediction model. The estimate $\epsilon_\theta(\mathbf{x}_t, t)$ only approximates the true noise ϵ , even though neural networks are expressive in principle under the *Universal Approximation Theorem* (Hornik et al., 1989; Nielsen, 2015; Yarotsky, 2022). Let $\Delta\epsilon_t(\mathbf{x}_t) = \epsilon - \epsilon_\theta(\mathbf{x}_t, t)$ denote this prediction error. By Eq. (5), any discrepancy in ϵ_θ induces a corresponding deviation in the predicted mean at step t :

$$\Delta\mu_t(\mathbf{x}_t) = -\frac{\beta_t}{\sqrt{1-\beta_t}} \frac{1}{\sqrt{1-\bar{\alpha}_t}} \Delta\epsilon_t(\mathbf{x}_t). \quad (\text{A.8})$$

Because each reverse step depends on the previous denoising state, this bias can accumulate along the trajectory and contribute to the final generation error.

A.2 PROOF OF THE TOTAL ACCUMULATED TRAJECTORY ERROR IN ODE-BASED SAMPLING

Building on Eq. (7), we illustrate the accumulated trajectory error using a one-step numerical solver. Let $\mathbf{x}^*(\tau)$ denote the ideal trajectory satisfying the ODE with the true score function $h^*(\mathbf{x}(\tau), \tau)$ and starting from a ground-truth initial condition $\mathbf{x}^*(T)$ drawn from the true diffused distribution $q_T(\mathbf{x}_T)$. Let $\hat{\mathbf{x}}_k$ denote the numerical trajectory produced by an ODE solver at discrete time steps $\tau_k = T - k\Delta t$ for $k = 0, 1, \dots, N$, and let $\hat{h}$ be the estimated model function. For a numerical trajectory from $\hat{\mathbf{x}}_0$ to $\hat{\mathbf{x}}_N$, we characterize the total accumulated trajectory error. Define $e_k = \hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)$ as the trajectory error at step k and consider the transition from τ_k to $\tau_{k+1} = \tau_k - \Delta t$.

Ideal path evolution. The variation-of-constants formula gives the ideal one-step evolution from τ_k down to τ_{k+1} :

$$\mathbf{x}^*(\tau_{k+1}) = \mathcal{G}(\tau_{k+1}, \tau_k) \mathbf{x}^*(\tau_k) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) h^*(\mathbf{x}^*(u), u) du, \quad (\text{A.9})$$

where $\mathcal{G}(t_2, t_1) = e^{\int_{t_1}^{t_2} f(v) dv}$ is the state transition operator for the linear part $\frac{d\mathbf{z}}{d\tau} = f(\tau)\mathbf{z}$. The integration limits reflect the backward-time denoising direction.

Numerical path evolution. A one-step numerical solver approximates this evolution rather than following the exact trajectory induced by $\hat{h}$ from $\hat{\mathbf{x}}_k$. Let $\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} \mid \hat{\mathbf{x}}_k, \hat{h})$ denote the exact solution at τ_{k+1} of the ODE using $\hat{h}$ and starting from $\hat{\mathbf{x}}_k$ at τ_k :

$$\hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} \mid \hat{\mathbf{x}}_k, \hat{h}) = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u \mid \hat{\mathbf{x}}_k, \hat{h}), u) du. \quad (\text{A.10})$$

The numerical method introduces the local truncation error δ_k :

$$\hat{\mathbf{x}}_{k+1} = \hat{\mathbf{x}}_{\text{exact}}(\tau_{k+1} \mid \hat{\mathbf{x}}_k, \hat{h}) + \delta_k. \quad (\text{A.11})$$

Thus,

$$\hat{\mathbf{x}}_{k+1} = \mathcal{G}(\tau_{k+1}, \tau_k) \hat{\mathbf{x}}_k + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u \mid \hat{\mathbf{x}}_k, \hat{h}), u) du + \delta_k. \quad (\text{A.12})$$

Error recurrence. Subtracting the ideal evolution in Eq. (A.9) from the numerical evolution in Eq. (A.12) gives:

$$\begin{aligned} e_{k+1} &= \hat{\mathbf{x}}_{k+1} - \mathbf{x}^*(\tau_{k+1}) \\ &= \mathcal{G}(\tau_{k+1}, \tau_k) (\hat{\mathbf{x}}_k - \mathbf{x}^*(\tau_k)) + \int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u \mid \hat{\mathbf{x}}_k, \hat{h}), u) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k. \end{aligned} \quad (\text{A.13})$$

Accumulated error formulation. This recurrence can be unrolled from $k = 0$ to $N - 1$. Let $\mathcal{G}(\tau_j, \tau_k) = \mathcal{G}(\tau_j, \tau_{j-1}), \dots, \mathcal{G}(\tau_{k+1}, \tau_k)$ for $j > k$, and $\mathcal{G}(\tau_k, \tau_k) = \mathbf{I}$. Repeated application of the

recurrence yields:

$$e_N = \mathcal{G}(\tau_N, \tau_0)e_0 + \sum_{k=0}^{N-1} \mathcal{G}(\tau_N, \tau_{k+1}) \cdot \left(\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h})) - h^*(\mathbf{x}^*(u), u) \right] du + \delta_k \right). \quad (\text{A.14})$$

Substituting $\tau_N = 0$, $\tau_0 = T$, $e_N = \hat{\mathbf{x}}_N - \mathbf{x}^*(0)$, and $e_0 = \hat{\mathbf{x}}_0 - \mathbf{x}^*(T)$ gives:

$$\hat{\mathbf{x}}_N - \mathbf{x}^*(0) = \underbrace{\mathcal{G}(0, T)(\hat{\mathbf{x}}_0 - \mathbf{x}^*(T))}_{\text{propagated initial error}} + \sum_{k=0}^{N-1} \mathcal{G}(0, \tau_{k+1}) \cdot \left(\underbrace{\int_{\tau_k}^{\tau_{k+1}} \mathcal{G}(\tau_{k+1}, u) \left[\hat{h}(\hat{\mathbf{x}}_{\text{exact}}(u | \hat{\mathbf{x}}_k, \hat{h}), u) - h^*(\mathbf{x}^*(u), u) \right] du}_{\text{single step model prediction error contribution}} + \underbrace{\delta_k}_{\text{single step truncation error}} \right). \quad (\text{A.15})$$

where the propagated initial error term captures the deterministic contribution of a particular initial error instance to the final path error, with its statistical origin in the diffused-prior gap.

A.3 THE DISCREPANCY BETWEEN DDPM AND DDIM IN THE FORWARD AND REVERSE PROCESS

As shown in Eq. (2), the DDPM (Ho et al., 2020) forward process can be written as:

$$\mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + (1 - \bar{\alpha}_t) \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{I}). \quad (\text{A.16})$$

The corresponding reverse process is defined as:

$$\mathbf{x}_{t-1}^{\text{DDPM}} = \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \frac{1 - \bar{\alpha}_t}{\sqrt{\bar{\alpha}_t} \sqrt{1 - \bar{\alpha}_t}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) + \sqrt{\beta_t} \boldsymbol{\epsilon}. \quad (\text{A.17})$$

For DDIM (Song et al., 2021), the forward process is identical to Eq. (A.16), while its deterministic reverse process with $\eta = 0$ is:

$$\begin{aligned} \mathbf{x}_{t-1}^{\text{DDIM}} &= \sqrt{\frac{\bar{\alpha}_{t-1}}{\bar{\alpha}_t}} (\mathbf{x}_t - \sqrt{1 - \bar{\alpha}_t} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t)) + \sqrt{1 - \bar{\alpha}_{t-1}} \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t) \\ &= \frac{1}{\sqrt{\bar{\alpha}_t}} \mathbf{x}_t - \left(\frac{\sqrt{1 - \bar{\alpha}_t}}{\sqrt{\bar{\alpha}_t}} - \sqrt{1 - \bar{\alpha}_{t-1}} \right) \boldsymbol{\epsilon}_\theta(\mathbf{x}_t, t). \end{aligned} \quad (\text{A.18})$$

Although DDPM and DDIM share the same forward diffusion process, DDIM is not obtained by simply removing the stochastic term from the DDPM reverse denoising process. The two methods use different coefficient formulations for the predicted noise term.

A.4 DATASET EXAMPLES AND DETAILS

Figure A1 presents representative examples from the three datasets used to instantiate our counting-hallucination protocol. As shown in Table A1, each dataset has a balanced distribution over its relevant categories, and Table A2 shows comparable distributions across object counts. These summaries indicate that the evaluated count constraints are not driven by obvious category or object-count imbalance. The datasets appear in the same order in both tables, and the category order in Table A1 is:

- **ToyShape:** triangle, square, pentagon;
- **SimObject:** mug, apple, clock;
- **RealHand:** finger.

Table A1: Object counts per category for each CountHalluSet dataset.

	ToyShape	SimObject	RealHand
Category 1	20,027	19,916	25,250
Category 2	19,902	20,101	-
Category 3	19,978	19,974	-

Table A2: Image counts grouped by total object count for each dataset.

	ToyShape	SimObject	RealHand
1 object	10,003	10,022	-
2 objects	10,087	9,965	-
3 objects	9,910	10,013	-
5 objects	-	-	25,250
Total	30,000	30,000	25,250

A.5 EXAMPLES OF COUNTING HALLUCINATIONS

We provide some representative examples of counting hallucinations corresponding to three datasets, as shown in Fig. A2. For ToyShape and SimObject, hallucinated samples either contain more than one instance of at least one category or contain no instances while retaining a normal background, as shown in Fig. A2(a) and Fig. A2(b). For RealHand, hallucinated samples are defined by incorrect valid-finger counts, such as 4 or 6 fingers, as shown in Fig. A2(c). The last two images in the first row of Fig. A2(c) appear at first to display five fingers, but one finger is partially incomplete and visibly artifacted, effectively leaving four valid fingers. They are therefore classified as counting hallucinations under the RealHand criterion.

A.6 EXAMPLES OF YOLO PREDICTIONS OF FINGERTIPS

Figure A3 presents representative YOLO predictions on generated RealHand images. The YOLO detector identifies fingertips across diverse poses and orientations, including both dorsal and palmar views. However, a count-based detector may not distinguish some anatomically implausible configurations, such as hands with duplicated thumbs (e.g., the penultimate sub-figure of the first row and the last sub-figure of the last row in Fig. A3(a)) or hands with five middle fingers but no thumb (e.g., the last sub-figure of the first row and the penultimate sub-figure of the last row in Fig. A3(a)). These samples are still detected as five-finger hands and therefore remain indistinguishable from anatomically valid ones under a pure finger-count criterion.

Figure A3(b) shows counting-hallucinated samples for which YOLO predicts incorrect fingertip counts, such as 3, 4, 6, or 7 fingertips. Figure A3(c) illustrates rare edge cases ($<1\%$) in which the detector struggles with unclear fingertip boundaries, cracked fingers, or floating artifacts. These rare cases can be partly mitigated by tuning YOLO inference hyperparameters, such as IoU and confidence thresholds, but doing so may reduce detection quality on the majority of counting-valid samples. Because the same detection setup is used consistently across all RealHand evaluations, these edge cases do not change the main conclusions, though they highlight a limitation of count-only automated assessment.

A.7 EXAMPLES OF THREE CLASSES OF GENERATED HAND IMAGES

Figure A4 shows representative examples of the three RealHand output classes: counting-valid, counting-hallucinated, and non-counting failure samples. As illustrated in Fig. 2, RealHand evaluation first applies the counting-ready indicator to separate countable samples, including counting-valid and counting-hallucinated images, from images unsuitable for counting. The unsuitable class typically contains severe visual artifacts, such as distorted, cracked, partially visible, or floating fingers. Counting-ready samples are then classified as counting-valid or counting-hallucinated according to whether the detected number of valid fingertips equals five.

A.8 EXTENDED DETAILS OF PTI-HAND EXPERIMENTS

In the PTI-Hand experiments, we use 10 hand-generation prompts for Stable-Diffusion-3.5-Medium to sample a range of prompt phrasings while preserving the same target content. The prompts are:

1. A single human hand, front view, all fingers in frame and visible.
2. A complete human hand, seen from the front, showing all fingers clearly, minimal background.
3. An isolated single hand, viewed from directly above, showing all fingers, simple studio background.
4. Overhead shot of a person’s hand, entire hand in frame, all fingers visible, plain background.
5. One hand, top view, five fingers visible, simple background.
6. A single hand, front or back view, flat against a surface, all fingers in frame and fully visible.
7. A single human hand, top-down view, all fingers visible, resting on a plain white surface.
8. A single human hand seen from above, fingers spread naturally, minimal background.
9. A human hand, top view, palm facing downward, all fingers clearly in frame, neutral background.
10. A single human hand captured from overhead, fingers extended, plain background with soft lighting.

Our definition of hallucination depends on identifying patterns absent from the relevant training distribution. For large pretrained diffusion models, the full training distribution is inaccessible, making this condition difficult to verify directly. To keep PTI-Hand quantification feasible and conservative, we treat any generated hand image with six or more fingers as counting-hallucinated. This criterion is conservative because real images can show fewer than five visible fingers due to occlusion, articulation, or viewpoint, whereas hands with six or more fingers are highly unlikely in real image data. Thus, finger counts ≥ 6 provide a reliable and empirically verifiable signal of counting hallucination in the pretrained text-to-image setting. Generated examples appear in Fig. A5.

A.9 EXTENDED DETAILS, RESULTS, AND DISCUSSION OF THE JOINT DIFFUSION MODEL

A.9.1 IMPLEMENTATION DETAILS IN TABLE 3

In the JDM experiments reported in Table 3, we fine-tune the denoising UNet for 120k steps, compared with 80k steps for the vanilla LDM. The increased budget is needed because extending the latent space from three to six channels introduces a mismatch with the pretrained LDM latent distribution. Additional optimization helps reduce this distributional gap and allows the joint image-mask modeling capacity to emerge.

A.9.2 FINE-TUNING PRETRAINED TEXT-TO-IMAGE MODELS WITH JOINT-DIFFUSION TRAINING

Table A3: Comparison of failure rates and image quality between fine-tuning SD-1.5 and fine-tuning SD-1.5 with the joint-diffusion training paradigm on RealHand. The first row reports the baseline obtained by fine-tuning Stable-Diffusion-1.5 on RealHand. The second row reports the constrained joint-diffusion variant, which freezes all intermediate layers of the baseline model and fine-tunes only the input and output layers with extended channel dimensions. Lower values indicate better performance.

Solver Name	Counting Hallucination Rate (CHR) ↓						Non-counting Failure Rate (NCFR) ↓						Fréchet Inception Distance (FID) ↓					
	DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2			DPM-Solver-1			DPM-Solver-2		
Sampling Steps	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100	25	50	100
SD-1.5	0.3362	0.3404	0.3630	0.3396	0.3376	0.3362	0.0059	0.0032	0.0020	0.0050	0.0026	0.0018	61.51	57.71	57.70	58.53	57.57	56.69
SD-1.5 with joint-diffusion finetuning	0.2838	0.3222	0.4145	0.3053	0.3277	0.3558	0.0947	0.0511	0.0487	0.0549	0.0362	0.0327	70.65	79.22	102.39	70.80	72.84	83.07

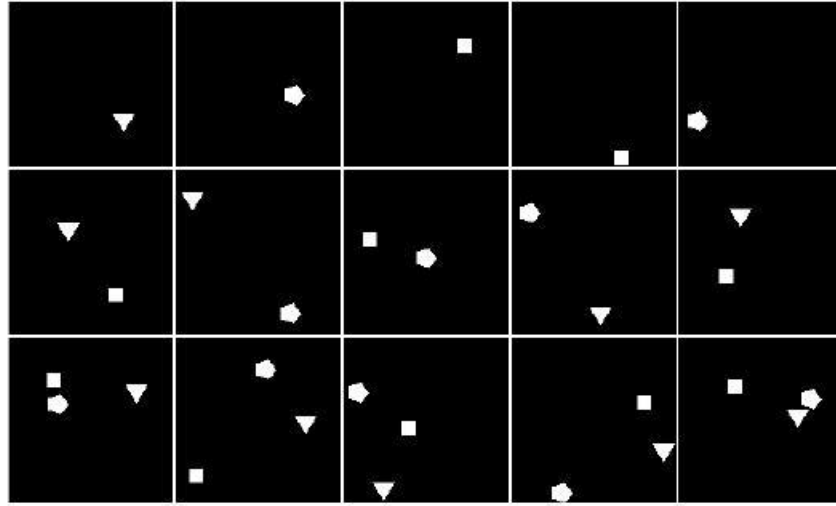
Importantly, JDM should be viewed as a **novel training paradigm** for improving factual generation in diffusion models rather than as a superior inference strategy. Because JDM only extends the

1026 input and output channels of the UNet while preserving the intermediate layer sizes, it **introduces a**
1027 **negligible inference-time overhead compared to vanilla LDM**. However, adding mask channels to
1028 a pretrained text-to-image model **disrupts the manifold geometry of the pretrained latent space**
1029 **and alters the original denoising distribution**, making direct finetuning of such a model nontrivial.

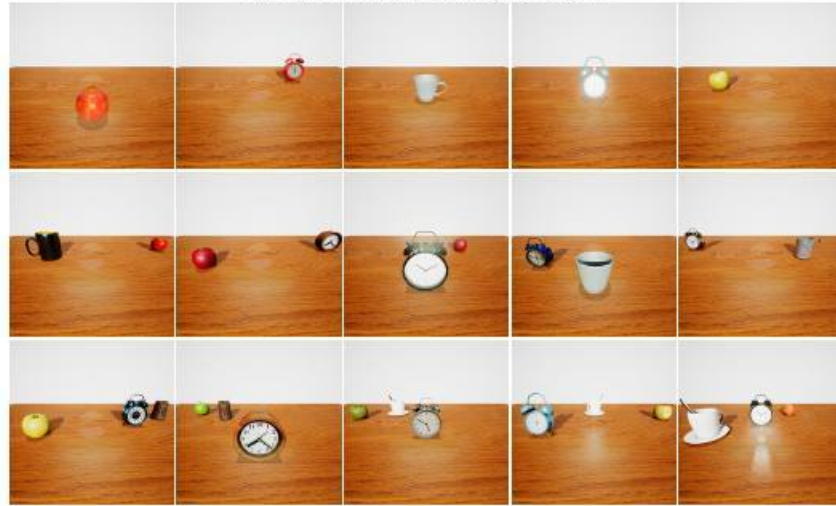
1030 Despite this challenge, JDM still produces promising reductions in counting hallucinations when
1031 applied to a large pretrained diffusion model under carefully controlled finetuning conditions. In
1032 practice, aligning the extended JDM latent space with the rigid latent geometry of a large pretrained
1033 model is difficult at the limited scale of RealHand. We therefore use a two-stage finetuning strategy:
1034 first fine-tune Stable-Diffusion-1.5 on RealHand, then freeze all intermediate UNet layers and fine-
1035 tune only the input and output layers with extended channels. As shown in Table A3, even this
1036 constrained setup yields lower counting hallucination rates at 25 and 50 sampling steps across both
1037 solvers.

1038 Overall, these results suggest that JDM is a promising direction for improving factual consistency
1039 when structural constraints can be represented alongside visual latents. Finally, **exploring its behavior**
1040 **and performance under true large-scale training remains an important and compelling avenue**
1041 **for future research**.

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(a) Representative ToyShape samples

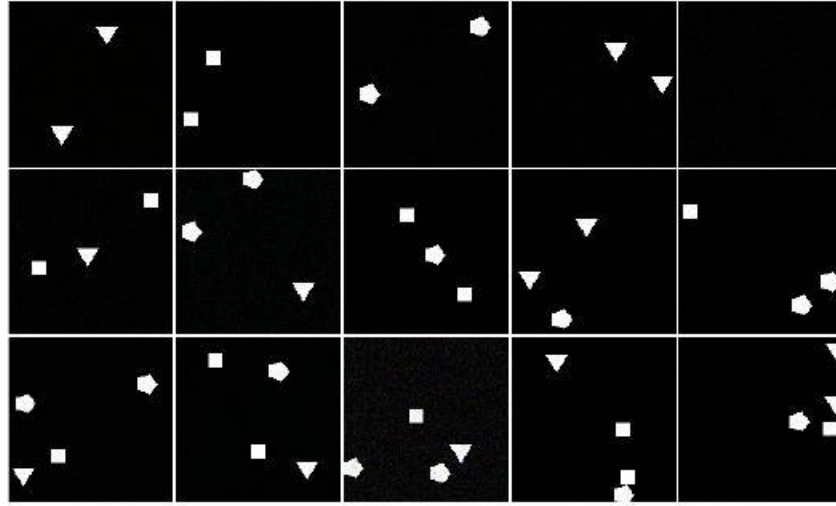


(b) Representative SimObject samples

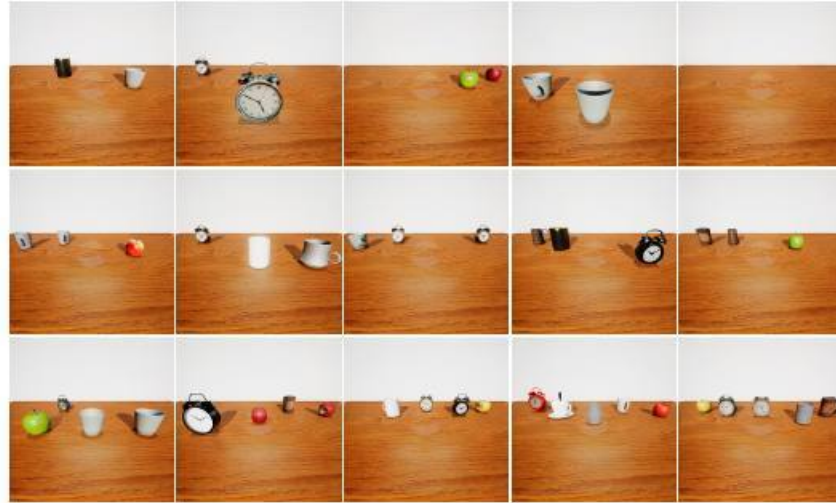


(c) Representative RealHand samples

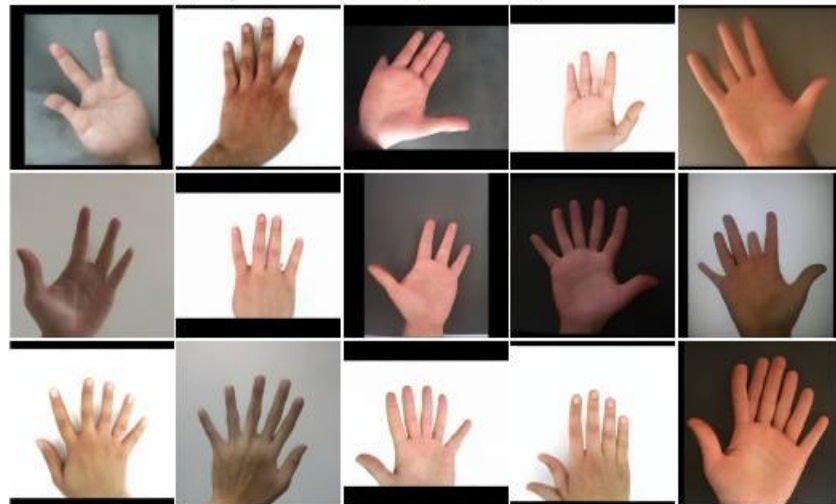
Figure A1: Representative samples from the CountHalluSet suite: ToyShape, SimObject, and RealHand.



(a) Representative ToyShape counting hallucinations

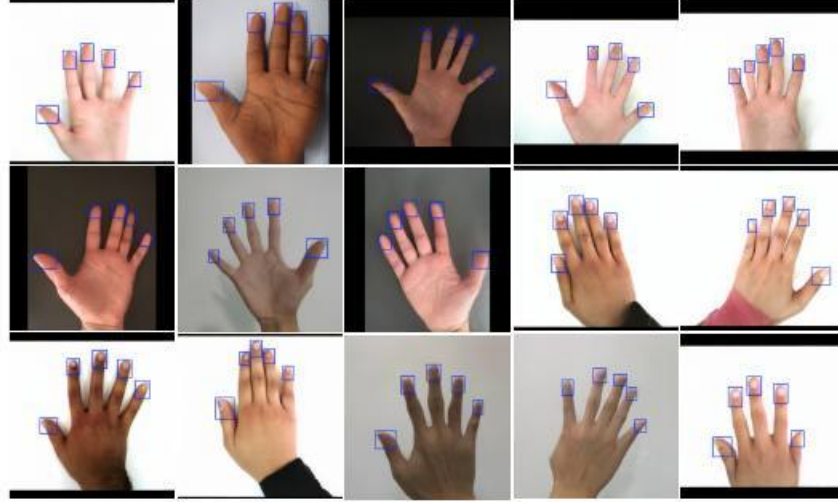


(b) Representative SimObject counting hallucinations

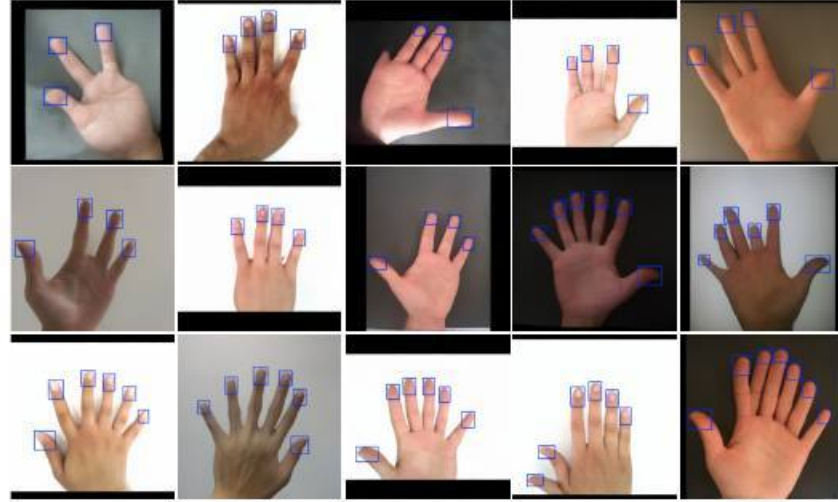


(c) Representative RealHand counting hallucinations

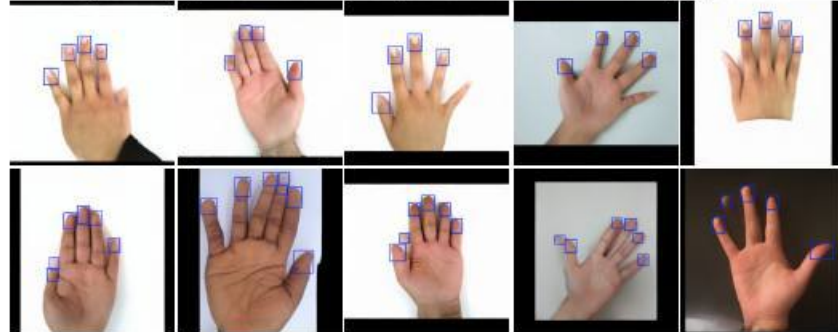
Figure A2: Representative counting hallucinations across ToyShape, SimObject, and RealHand.



(a) Counting-valid YOLO predictions with five detected fingers

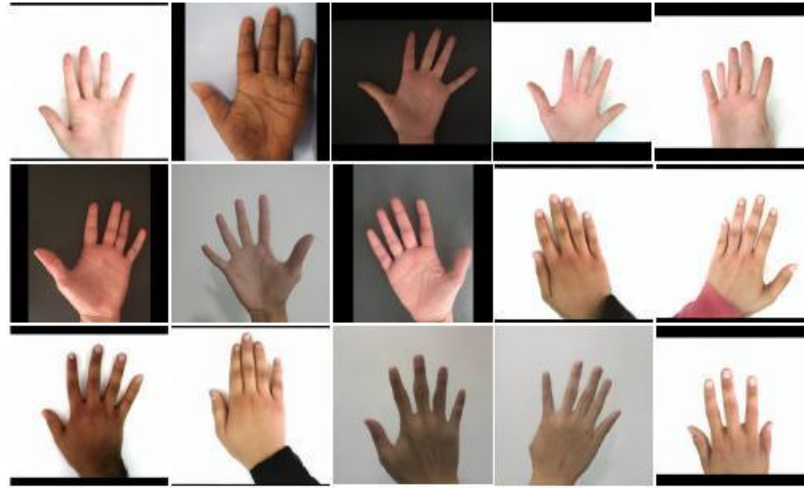


(b) Counting-hallucinated YOLO predictions with non-five fingertip counts

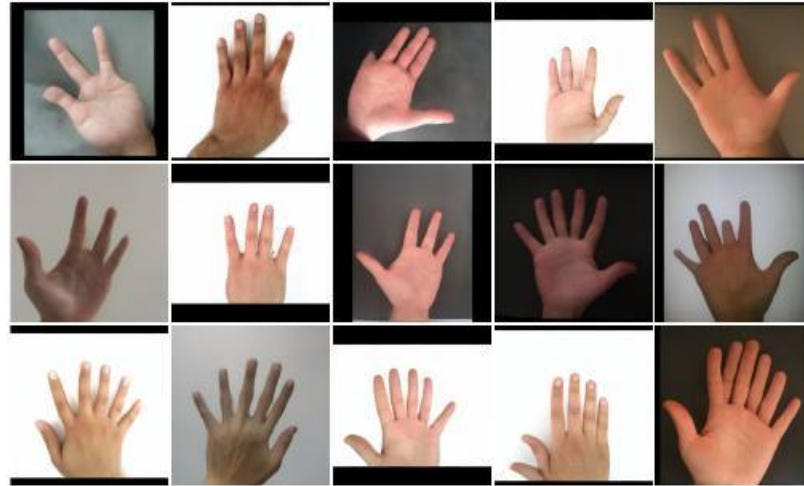


(c) Rare edge cases for YOLO fingertip prediction

Figure A3: YOLO predictions on generated RealHand images. (a) Counting-valid samples with five fingertip bounding boxes predicted by YOLO, including cases with two thumbs. (b) Counting-hallucinated samples with incorrect fingertip count predictions, such as 3, 4, 6, or 7 detected fingertips. (c) Rare edge cases that the counting-ready indicator does not filter out and that YOLO struggles to handle, including blurred, incomplete, or anatomically disconnected fingers.



(a) Counting-valid samples with five valid fingers



(b) Counting-hallucinated samples with non-five valid-finger counts



(c) Non-counting failure samples unsuitable for counting

Figure A4: Three classes of generated RealHand images. (a) Counting-valid samples contain five clearly discernible fingers. (b) Counting-hallucinated samples contain an incorrect number of valid fingers, such as 4 or 6. (c) Non-counting failures are unsuitable for counting because of severe visual artifacts, such as distorted, cracked, partially visible, or floating fingers.



(a) Counting-valid samples with five valid fingers



(b) Counting-hallucinated samples with more than 5 fingers



(c) Non-counting failure samples unsuitable for counting

Figure A5: Three classes of hand images generated by pretrained Stable-Diffusion-3.5-Medium (PTI-Hand). (a) Counting-valid samples contain five clearly discernible fingers. (b) Counting-hallucinated samples contain more than five valid fingers, such as 6 or 7. (c) Non-counting failures are unsuitable for counting because of severe visual artifacts, such as distorted, cracked, partially visible, or floating fingers.